

User Manual

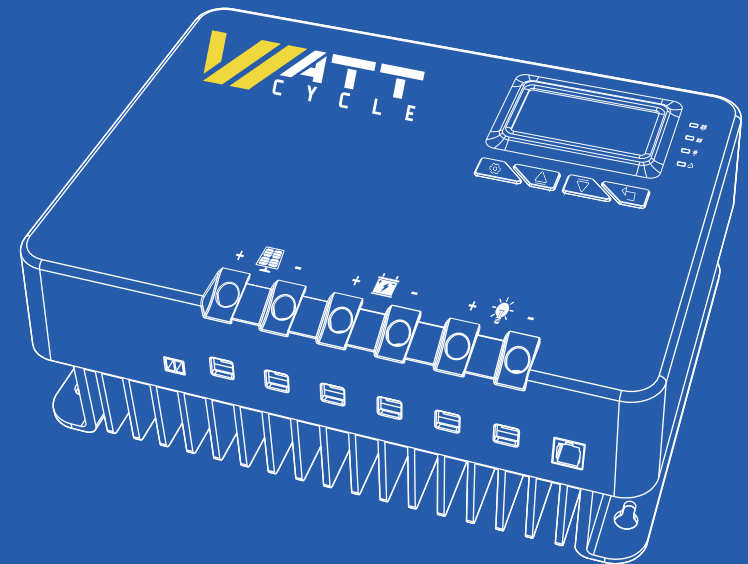
MPPT Solar Charge Controller



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Email: service@wattcycle.com

1. Warnings and Tools Icon Chart

Icons	Name	Meaning
	Danger of electric shock warning	Do not touch the electric connections, there is danger of electrical shock.
	High surface warming	Do not touch the surface of the appliance, while it is running, it will get hot.
	Environmental Hazard	Electronic Equipment. Do not put in landfill.
	Wire Cutter	A wire cutter is needed for cutting and stripping prior wires for connection.
	Multi-meter	A multi-meter is needed for testing equipment and verifying polarity of cables.
	Anti-static Glove	Anti-static gloves are recommended to prevent controller damage caused by static electricity.
	Insulating Tape	Insulating tape is recommended to safely insulate spliced or bare wires.
	Screwdriver	A common size screwdriver is needed to attach wires to the controller.

2. Safety Tips

Read this manual carefully. It contains important instructions that need to be followed during installation, operation and maintenance.

Save these instructions for future reference on operation and maintenance.

- Ensure that the equipment is used under the correct operating conditions. Never operate it in a wet environment. And please don't plug in any AC source to this DC-DC product, or it may cause a fire or burn the device.

- Lithium batteries are not allowed to be reverse connected, for risks may occur if the wires are connected in reverse when the battery is being activated.

- Don't plug the power inverter directly into the controller DC load output, otherwise it might cause short-circuit to this device or even fire due to over-current.

- Always keep children away from this device.

- Please don't keep the battery side open for too long while the PV input keeps plugging in, or it may cause a screen failure of this device.

- Solar panels can generate high voltages and currents, so protect the solar modules from incident light during installation, e.g. cover them. It is recommended that installation should be performed by a qualified electrician.

- To avoid damage to the battery or controller, use proper fuses or DC breakers when wiring. Please do not hesitate to contact the professionals if you need help with fuse and DC breaker sizing.

- Connecting wires to this device can generate sparks, please use only insulated tools.

- Be certain to use the correct gauge of wire, see the table below of recommended wire size for various current loads.

Solar Input Current	5A	10A	20A	30A	40A
Wire Cross Section Area (mm ²)	1.5	2.5	5	8	10
Wire AWG	15	13	10	8	7

* It is recommended that the current should not exceed 5A per square millimeter.

3. Product Features

Thank you for choosing our products. This MPPT solar charge controller is a device for solar charge regulation and direct current output load control. This device is mainly used in small and medium-sized off-grid solar power systems.

The MPPT charge controllers' features are as follows:

·Ultra fast MPP tracking

The solar charger contains an ultra fast MPPT controller. This is especially beneficial when the solar light intensity is constantly changing, as is the case during cloudy weather. Because of the ultra fast MPPT controller, 30% more energy is harvested compared to solar chargers with a PWM controller and up to 10% more compared to slower MPPT controllers.

·Optimal solar yield

The solar charger has an innovative tracking algorithm. It will always maximize energy harvest by locking to the optimum MPP (Maximum Power Point). If partial shading occurs, two or more maximum power points may be present on the power-voltage curve. Conventional MPPTs tend to lock to a local MPP, which may not be the optimum MPP.

·Built-in buffer allows max 25% exceeding rated power input.

·Charging modes are available for most common deep-cycle battery types in the market, including AGM (sealed lead acid batteries), GEL, Flooded, and Lithium. Use mode is also available for customized parameters setting.

·Auto recognition of 12V/24V battery system voltage is available to M2420/M2430/M2440 models (lithium battery excluded).

·Provides multiple load control mode options for light based, time based and manually adjusted scenarios and low no-load loss.

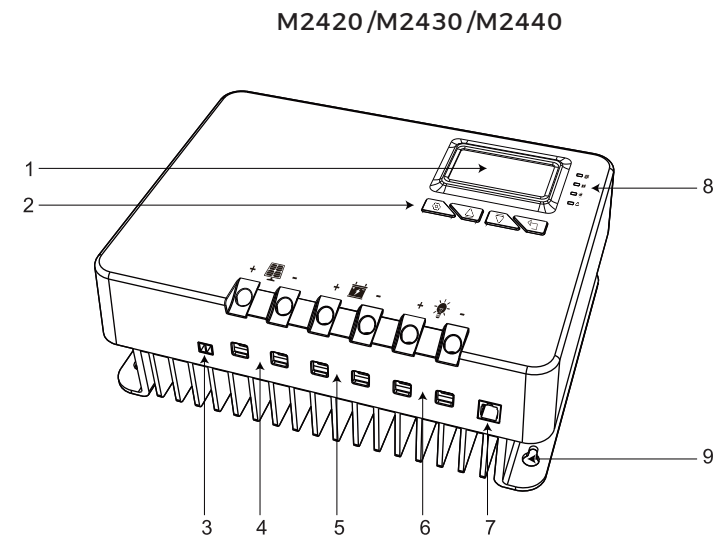
·It has industrial grade design with full range of electronic protections, such as battery over-charge, battery over-discharge, PV over-voltage, controller over-heating, DC load short-circuit, DC load over-load, as well as reverse polarity protection for PV and battery sides, etc.

·Supports RS485 communication with Modbus protocol (RJ12).

·Supports parallel charge function of dual units by linking each other with the parallel charge cable (optional accessory, not in the standard package list).

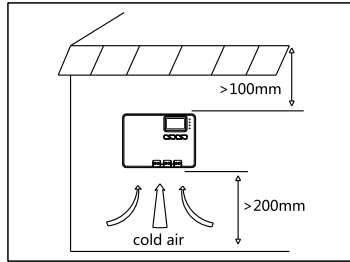
·Supports remote display screen for monitoring and operation, linking through RJ12 port in the controller (optional accessory, not in the standard package list).

4. Device Diagram



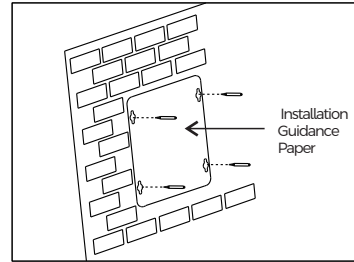
#	Description	#	Description
1	LCD Display Screen	6	DC Load Terminals
2	Function Key ([SET], [UP], [DOWN], [ESC])	7	RS485 Communication Port (RJ12)
3	External Temperature Sensor Terminal	8	LED Indicator (PV, BAT, LOAD, FAULT)
4	Solar Terminals	9	Installation Mounting Holes
5	Battery Terminals		

5. Mounting Instruction



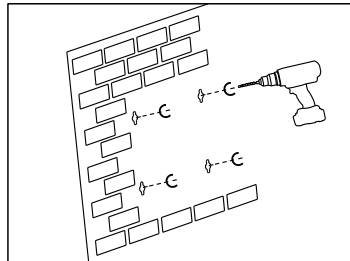
Step 1

Find a safe place to install under cool and dry weather conditions.



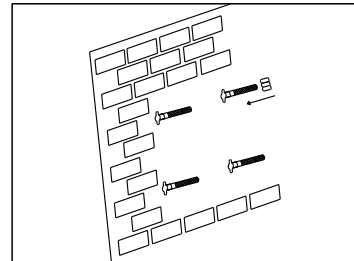
Step 2

Mark 4 spots on the mounting surface through the 4 pilot holes on the pilot paper as installation holes.



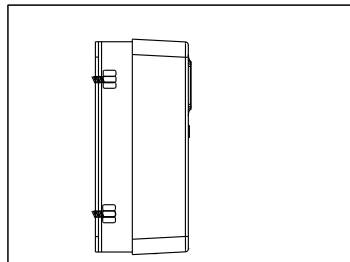
Step 3

Drill holes at the marked spots.



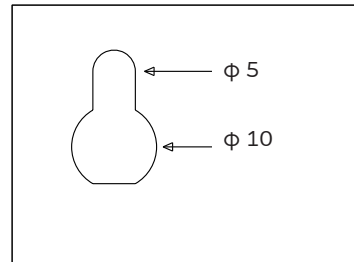
Step 4

Insert pilot screws in the mounting hole.



Step 5

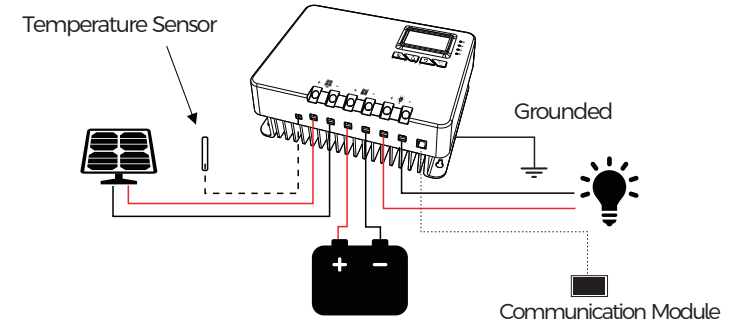
Fasten the controller into the pilot screws.



Step 6

Wire battery, solar, DC load and other accessories to the controller.

6. Wire Connection Sequences



Please connect the wires in the following order when installing your MPPT controller:

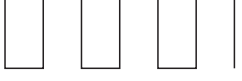
- Connect the positive battery wire followed by the negative battery wire.
- Make sure your solar panels are fully covered to prevent electrical shock.
- Connect the positive solar array output wire followed by the negative solar array output wire.
- Connect DC load wires to the DC load output (if applicable).
- Connect the external temperature sensor to its terminal shown above.

7. LED Light Signal Interpretation Chart

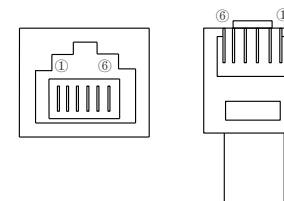
LED Name	LED Display	Signal Indication
PV	Off	Solar Input Not Charging *PV LED is generally off during nighttime.
	Double Flash	Solar Input Detected / Solar Over Voltage
	Single Flash	Solar Input Reverse Polarity
	Steady On	MPPT Charge Mode
	Slow Flash	In Equalize/Boost/Float Charge
BATTERY	Single Flash	Battery Input Reverse Polarity
	Fast Flash	Battery Over Voltage
	Slow Flash	Battery Over Discharged
	Steady On	Battery On
LOAD	Off	Load Off
	Fast Flash	DC Load Short Circuit / Overload
	Steady On	DC Load On
FAULT	Off	No Errors
	Steady On	System Error – Check Error Code

*Check the Fault light to spot if a system error may be present.

8. LED Flash Rhythm Chart

Flash Status	Indication	Description
Steady On	on off 	LED light on.
Off	on off 	LED light off.
Fast Flash	on off 	LED light blinks at a frequency of 2Hz (twice every second).
Slow Flash	on off 	LED light blinks at a frequency of 0.5Hz (once every two seconds).
Single Flash	on off 	LED light blinks for 0.1second after every 2 seconds.
Double Flash	on off 	LED light blinks for 0.1second twice after every 4 seconds.

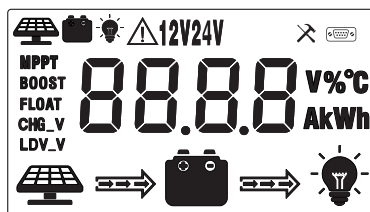
9. RS485 Connection Diagram (RJ12)



RS485 PIN (RJ12)					
PIN-1	PIN-2	PIN-3	PIN-4	PIN-5	PIN-6
VDD	VDD	GND	GND	D-	D+

*Support 3.3V, 20mA

10. LCD Display Interface Overview



Display Section	Display Layout
Charge and Load Status	  
Charge & Set Mode	MPPT BOOST FLOAT CHG_V LDV_V
Parameter	88.8.8
Unit	V%°C AkWh
Active Functions	    12V24V  

11. LCD Status Information

Status Icon	Indication	Status	Description
	Solar Charge Indication	Flowing	Solar Power Charging Battery
		Off	Solar Power Not Charging Battery
	DC Load Indication	Flowing	DC Load Drawing Power
		Off	DC Load Off
MPPT	Charge Mode	Steady On	MPPT Charge Mode
BOOST			Boost Charge Mode
FLOAT			Float Charge Mode

Status Icon	Indication	Status	Description
CHG_V	Voltage Setting (Lithium battery)	On	Setting Charge Voltage
		Off	Charge Voltage Has Been Set
LDV_V	Over Discharge Volt Settings (Lithium battery)	On	Setting Discharge Voltage
		Off	Discharge Voltage Has Been Set
12V24V	System Volt	One of 12V/24V ON	Current System Voltage
	Warning	On	System failure occur
		Off	No System failure
	Setting	On	Setting Mode
		Off	View Mode
	Communication	On	In the Modbus communication
		Off	No communication
	Solar Icon	Steady On	Daylight Detected
		Off	No Daylight Detected
		Fast Flash	Solar panel Over Voltage
	Battery Icon	Steady On	Battery Connected and Functional
		Off	No Battery Connection
		Fast Flash	Over Voltage
		Slow Flash	Battery Over-Discharged or Reverse Connection
	Load Status	ON	Load On
		OFF	Load Off
		Fast Flash	DC Load Short Circuit
		Slow Flash	Over Load

*There is no charge mode displayed on the screen of the controller when it's in equalize charge mode.

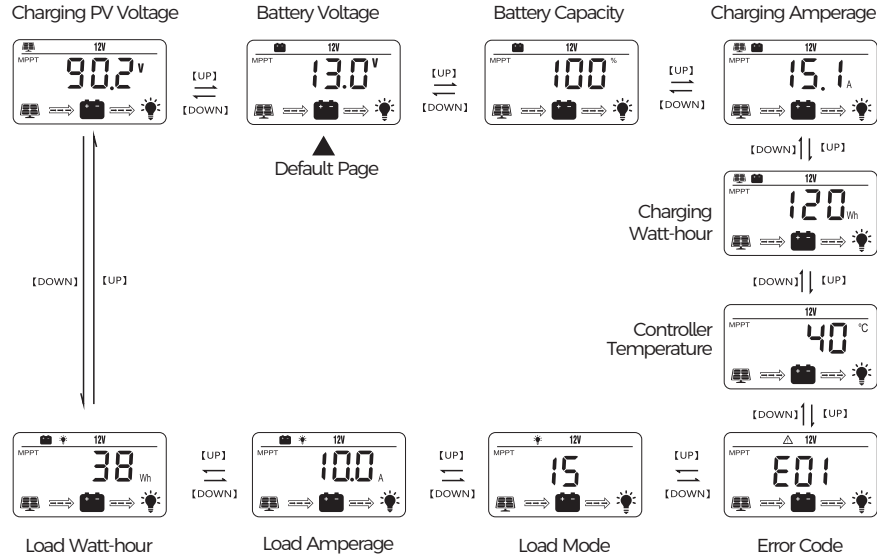
12. Key Functionality Chart

Function Key	System Mode	Input	Input Function
	View Mode	Long Press	Enter SET mode
	View Mode	Short Press	View Previous Page
	View Mode	Short Press	View Next Page
	View Mode	Short Press	DC Load On/Off (Manual Control Program Only)

Function Key	System Mode	Input	Input Function
	SET Mode	Long Press	Save Data & Exit SET Mode
		Short Press	Next Setting Item
	SET Mode	Short Press	Increase Parameter Value
	SET Mode	Short Press	Decrease Parameter Value
	SET Mode	Short Press	Exit SET Mode Without Saving

13. LCD Display Rules & Cycles

LCD Screen Display Cycle



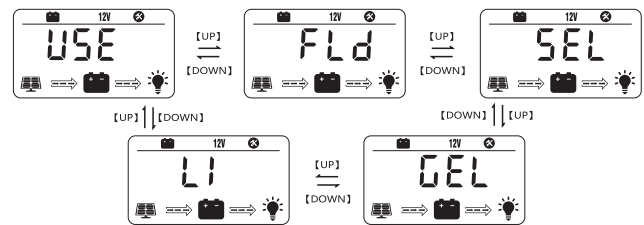
The battery voltage view will be displayed by default. Use the up and down arrow keys to cycle through different views. The battery voltage view will resume upon 30seconds of inactivity. The error code view will be displayed when an error is detected. The backlight on the screen will be on for 20 seconds with any button operation.

14. Key Functionality Chart

Setting Battery Mode

Enter SET mode by long pressing the SET key in any view page other than Load Mode.

Use the up and down arrow keys to select battery mode, then long press SET key to save.

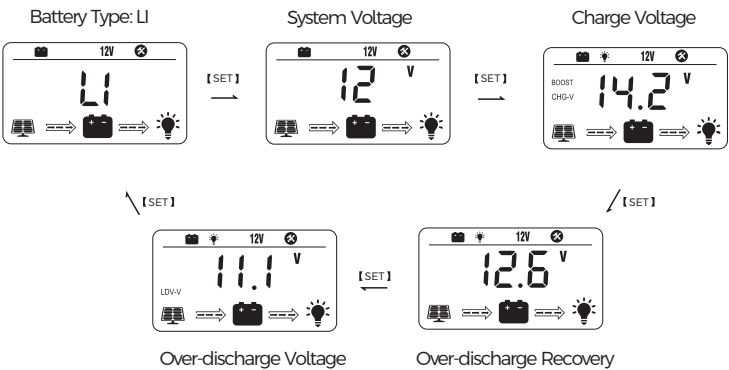


Abbreviations	Battery Types	Description
FLD	Flooded Battery	Auto-recognition with default parameters set for each type of battery.
SEL	Sealed/AGM Battery	
GEL	Gel Battery	
LI	Lithium Battery	Some parameters can be customized.
USE	Advanced User Mode	Most parameters can be customized.

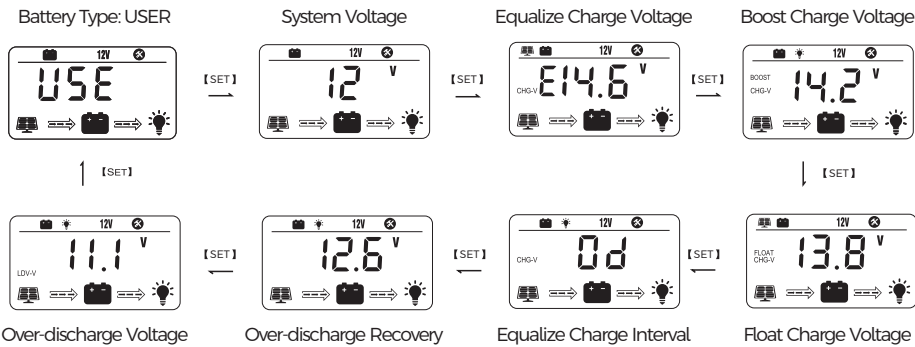
Advanced Battery Settings

In Lithium or User mode, short press the SET key to cycle through each parameter view. Use the up and down arrow key to adjust parameter value, then long press SET key to save.

For Battery Type: LI



For Battery Type: USE



Load Mode Settings

Enter Load SET Mode by long pressing the SET key in Load Mode view only.

Use the arrow key to cycle through load modes before long pressing SET to save and exit. Short pressing ESC will exit without saving.



Mode	Definition	Description
0	Daylight Auto-Control	DC load turns on when no daylight is detected.
1-14	Daylight On/Timer Off	DC load turns on when no daylight is detected. DC load turns off according to the timer.
15	Manual Mode	DC load turns on/off by pressing the Return key.
16	Testing Mode	DC load turns on and off in quick succession.
17	Always On	DC load stays on.

*For load mode 1-14, the number means the load lasting time, e.g, "1" means the load would turn off in 1 hour after turning on, "8" means the load will turn off in 8 hours. Please notice that the sunlight would turn off the load for all load modes 1-14, even the timer hasn't run out yet.

15. Error Code Chart

An error code will be displayed in the following ways if an error occurs:

Code	Error	Description & Quick Troubleshoot
E00	No Error	No action needed.
E01	Battery Over-discharged	Battery voltage is too low. DC load will be turned off until battery re-charges to recovery voltage.
E02	Battery Over-voltage	Battery voltage has exceeded controller's limit. Check battery bank voltage for compatibility with controller.
E04	Load Short Circuit	DC load short circuit.
E05	Load Overload	DC load power draw exceeds controller capability. Reduce load size or upgrade to a higher load capacity controller.
E06	Overheating	Controller exceeds ambient temperature limit. Ensure the controller is placed in a well-ventilated cool, dry place.
E07	Environmental Overtemperature	The environment temperature sampled by external temperature probe is too high.
E10	Solar Over-voltage	Solar array voltage exceeds controller rated input voltage. Decrease the voltage of solar panels connected to the controller.
E13	Solar Reverse Polarity	Solar array input wires connected with reverse polarity. Disconnect and re-connect with correct wire polarity.
E14	Battery Reverse Polarity	Battery connection wires connected with reverse polarity. Disconnect and re-connect with correct wire polarity.

*Please contact professions for technical support for additional troubleshooting.

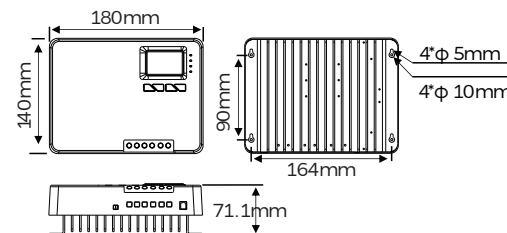
*Controller M4860 would not display error code "E13" on the screen.

16. Controller Specification

The variable "n" is adopted as a multiplying factor when calculating parameter voltages, the rule for "n" is listed as: if battery system voltage is 12V, n=1; 24V, n=2.
For example, the equalize charge voltage for a 12V FLD (Flooded) battery bank is $14.8V \times 1 = 14.8V$. The equalizing charge voltage for a 24V FLD (Flooded) battery bank is $14.8V \times 2 = 29.6V$.

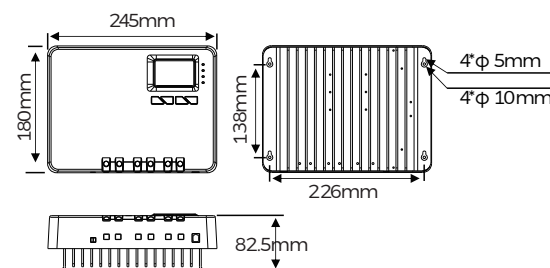
Parameter	Value				
Model No.	M2420	M2430	M2440		
Battery System Voltage	12V/24V Auto (FLD/GEL/SEL/USE) Manual (LI/USE)				
No-load Loss	12ma(12V),10ma(24V)				
Max Solar Input Voltage	<100Voc				
Rated Solar Charge Current	20 A	30 A	40 A		
Max Solar Input Power	300W/12V 600W/24V	450W/12V 900W/24V	600W/12V 1200W/24V		
Light Control Voltage	5V*n				
Light Control Delay Time	10 s				
Max Load Output Current	20A				
Operating Temperature	-35°C ~ +45°C				
IP Protection	IP32				
Net Weight	1.21kg	2.0kg	2.0kg		
Communication Port	RS485 (RJ12)				
Operating Altitude	≤3000 meters				
Controller Dimension(mm)	180*140*71.1	245*180*82.5			
Parameter	Battery Parameters				
Battery Types	FLD	SEL	GEL(default)	USE(adjustable)	LI (adjustable)
Equalize Charge Voltage	14.8V*n	14.6V*n	—	Default	--
Boost Charge Voltage	14.6V*n	14.4V*n	14.2V*n	Default: GEL	Default:14.2V*n
Float Charge Voltage	13.8V*n			Default: GEL	--
Boost Charge Recovery Voltage	13.2V*n			Default: GEL	--
Over-discharge Recovery Voltage	12.6V*n			Default: GEL	--
Over-discharge Voltage	11.1V*n			Default: GEL	Default :11.1V*n
Auto Temperature Compensation	-3m V/2V/°C			Default: GEL	--

17. Product Dimension



M2420 Model

Product Dimension: 180*140*71.1mm
Installation Area Dimension: 164*90mm
Installation Hole Size: φ5mm & φ10mm
Connection Socket Size: 7.5*10mm



M2430 & M2440 Model

Product Dimension: 245*180*82.5mm
Installation Area Dimension: 226*138mm
Installation Hole Size: φ5mm & φ10mm
Connection Socket Size: 7.5*10mm